Rockchip MIPI DSI 接口多供屏兼容开发指南

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产品版本

芯片名称	**u-boot**版本
RK3288	next-dev
RK3399	next-dev
RK3326/PX30	next-dev
RK356x	next-dev
RK3588	next-dev
RK3576	next-dev

读者对象

本文档（本指南）主要适用于以下工程师：

技术支持工程师

软件开发工程师

修订记录

版本号	作者	修改日期	修改说明
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目录

Rockchip MIPI DSI 接口多供屏兼容开发指南
1. Introduction
2. 核心功能说明
   2.1 多屏兼容机制
      2.1.1 动态面板检测
      2.1.2 设备树热修复
   2.2 设备树配置
      2.2.1 多面板节点定义
      2.2.2 面板 ID 配置
   2.3 驱动和设备树联动流程
      2.3.1 屏幕检测与切换流程
      2.3.2 完整补丁
3. 调试
   3.1 关键日志分析
   3.2 FDT调试

1. Introduction

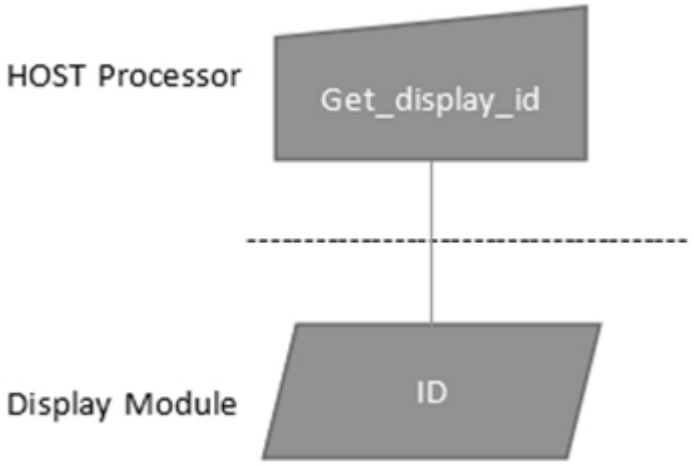
本指南旨在帮助开发人员和使用者在 Rockchip 含有 MIPI DSI 接口的 SoC 平台上配置和启用对多个二供 MIPI 显示屏的兼容支持。二供屏指的是项目中备用的不同供应商提供的显示屏，本指南将指导如何通过驱动层适配与设备树配置来实现同一个主板硬件环境搭配这些二供屏显示屏在同一个固件下能够动态检测切换并显示。

2. 核心功能说明

2.1 多屏兼容机制

2.1.1 动态面板检测

通过 MIPI DSI Lane0 双向传输的功能特性，读取对端屏幕的 ID 信息，自动匹配预定义的显示屏设备树配置描述。



2.1.2 设备树热修复

在 u-boot 阶段的显示系统使能显示通路之前，先获取对端显示屏的 ID，根据匹配结果动态调整设备树节点状态（"status"）,仅激活匹配的显示屏设备树配置（初始化序列、分辨率信息等）描述。

2.2 设备树配置

2.2.1 多面板节点定义

在 DTS 中对所有的二供屏描述上电控制供电使能、RST、背光、MIPI DSI 工作模式、通道数、是否使能 DSC、DPHY 或 CPHY、屏初始化序列和对应屏分辨率等信息。

```
&dsi0 {
    status = "okay";

    dsi0_panel0: panel0@0 {
        status = "okay";
        compatible = "simple-panel-dsi";
        reg = <0>;

        dsi,flags = <(MIPI_DSI_MODE_LPM | MIPI_DSI_MODE_EOT_PACKET)>;
        dsi,format = <MIPI_DSI_FMT_RGB888>;
        dsi,lanes  = <4>;

        slice-width = <720>;
        slice-height = <65>;
        version-major = <1>;
        version-minor = <1>;
        compressed-data;

        id-reg = <0xdb>;
        id = [10];

        panel-init-sequence = [
            /* panel0 init sequence */
        ];

        panel-exit-sequence = [
            /* panel0 exit sequence */
        ];

        disp0_timings0: display-timings {
            native-mode = <&dsi_timing0>;
            dsi_timing0: timing0 {
                clock-frequency = <280000000>;
                hactive = <1440>;
                vactive = <3120>;
                hfront-porch = <16>;
                hsync-len = <8>;
                hback-porch = <8>;
                vfront-porch = <4>;
                vsync-len = <16>;
                vback-porch = <8>;
                hsync-active = <0>;
                vsync-active = <0>;
                de-active = <0>;
```

```
                pixelclk-active = <0>;
            };
        };

        ports {
            #address-cells = <1>;
            #size-cells = <0>;

            port@0 {
                reg = <0>;
                panel0_in_dsi0: endpoint {
                    remote-endpoint = <&dsi0_out_panel>;
                };
            };
        };
    };

    dsi0_panel1: panel1@0 {
        status = "okay";
        compatible = "simple-panel-dsi";
        reg = <0>;

        dsi,flags = <(MIPI_DSI_MODE_VIDEO | MIPI_DSI_MODE_VIDEO_BURST |
            MIPI_DSI_MODE_LPM | MIPI_DSI_MODE_EOT_PACKET)>;
        dsi,format = <MIPI_DSI_FMT_RGB888>;
        dsi,lanes  = <2>;

        id-reg = <0xdb>;
        id = [72];

        panel-init-sequence = [
            /* panel1 init sequence */
        ];

        panel-exit-sequence = [
            /* panel1 exit sequence */
        ];

        disp0_timings1: display-timings {
            native-mode = <&dsi_timing1>;
            dsi_timing1: timing0 {
                clock-frequency = <110000000>;
                hactive = <1080>;
                vactive = <1920>;
                hfront-porch = <15>;
                hsync-len = <2>;
                hback-porch = <30>;
                vfront-porch = <15>;
                vsync-len = <2>;
                vback-porch = <15>;
                hsync-active = <0>;
                vsync-active = <0>;
                de-active = <0>;
```

```
                pixelclk-active = <1>;
            };
        };

        ports {
            #address-cells = <1>;
            #size-cells = <0>;

            port@0 {
                reg = <0>;
                panel1_in_dsi0: endpoint {
                    remote-endpoint = <&dsi0_out_panel>;
                };
            };
        };
    };

    /* more panel */

    ports {
        #address-cells = <1>;
        #size-cells = <0>;

        port@1 {
            reg = <1>;
            dsi0_out_panel: endpoint {
                remote-endpoint = <&panel0_in_dsi0>;
            };
        };
    };

};
```

2.2.2 面板 ID 配置

先向屏厂确认获取各个二供屏 ID 的命令和预期ID，并实际在屏端实践下是否能够按预期成功获取，然后将 ID 参数描述到 DTS 中。

属性	作用	示例值
id-reg	存储屏端 ID 寄存器地址	<0xdb> (寄存器 0xdb，不同屏地址一般不同)
id	预期 ID 值（十六进制数组）	[10] (长度也可能根据屏有差异)

2.3 驱动和设备树联动流程

2.3.1 屏幕检测与切换流程

```
开始
  |
  v
[驱动加载]
  |
  v
[遍历 DSI 控制器下所有 panel 节点，存储对应的信息]
  |
  v
[预初始化（pre_prepare）]
  |
  |-- [初始化 MIPI DSI]
  |
  v
[ID 匹配阶段]
  |
  |-- [发送 panel-read-id-sequence（若定义）]
  |-- [通过 DCS 读取 id-reg 的值]
  |-- [与设备树中 id 值比对]
  |-- [记录匹配的面板索引]
  |
  v
[动态设备树修复（fixup_dts）]
  |
  |-- [激活匹配面板的 remote-endpoint]
  |-- [禁用未匹配面板节点：fdt_setprop_string(..., "status", "disabled")]
  |
  v
[显示初始化]
  |
  |-- [使用匹配面板的时序和命令序列初始化]
  |
  v
结束
```

由于多屏兼容逻辑是在 u-boot 阶段事先获取屏端 ID 并遍历设备树中 &dsi 节点下的所有 panel 节点，因此对开机时间有一定的开销(主要是屏端的上电时序延迟)，并且多屏兼容依赖 u-boot 开机 LOGO 功能。

2.3.2 完整补丁

```
From ace0a1101133a65dd535043437747916c3f331dc Mon Sep 17 00:00:00 2001
From: Zhibin Huang <zhibin.huang@rock-chips.com>
Date: Sat, 23 Mar 2024 21:39:32 +0800
Subject: [PATCH] video/drm: support multi mipi screen
```

```
Type: Function
Redmine ID: N/A
Associated modifications: N/A
Test: N/A

Signed-off-by: Zhibin Huang <zhibin.huang@rock-chips.com>
Change-Id: I4abe3e8bf6a4b226bc400054fa1c0ba523aafd93
---

diff --git a/drivers/video/drm/dw_mipi_dsi.c b/drivers/video/drm/dw_mipi_dsi.c
index 2ebe826..6e13b24 100644
--- a/drivers/video/drm/dw_mipi_dsi.c
+++ b/drivers/video/drm/dw_mipi_dsi.c
@@ -18,6 +18,7 @@
 #include <syscon.h>
 #include <asm/arch-rockchip/clock.h>
 #include <linux/iopoll.h>
+#include <fdt_support.h>

 #include "rockchip_display.h"
 #include "rockchip_crtc.h"
@@ -1259,6 +1260,46 @@
        dw_mipi_dsi_pre_enable(dsi->slave);
 }

+static void dw_mipi_dsi_connector_pre_prepare(struct rockchip_connector *conn,
+                         struct display_state *state)
+{
+   struct connector_state *conn_state = &state->conn_state;
+   struct dw_mipi_dsi *dsi = dev_get_priv(conn->dev);
+   unsigned long lane_rate = 1000000000UL;
+
+   memcpy(&dsi->mode, &conn_state->mode, sizeof(struct drm_display_mode));
+   if (dsi->slave)
+       memcpy(&dsi->slave->mode, &dsi->mode,
+              sizeof(struct drm_display_mode));
+
+   if (dsi->dphy.phy)
+       dw_mipi_dsi_set_hs_clk(dsi, lane_rate);
+   else
+       dw_mipi_dsi_set_pll(dsi, lane_rate);
+
+   if (dsi->slave && dsi->slave->dphy.phy)
+       dw_mipi_dsi_set_hs_clk(dsi->slave, lane_rate);
+
+   printf("final DSI-Link bandwidth: %u Mbps x %d\n",
+          dsi->lane_mbps, dsi->slave ? dsi->lanes * 2 : dsi->lanes);
+
+   dw_mipi_dsi_init(dsi);
+   dw_mipi_dsi_dpi_config(dsi, &dsi->mode);
+   dw_mipi_dsi_packet_handler_config(dsi);
+   dw_mipi_dsi_video_mode_config(dsi);
```

```
+	dw_mipi_dsi_command_mode_config(dsi);
+	dsi_update_bits(dsi, DSI_MODE_CFG, CMD_VIDEO_MODE, COMMAND_MODE);
+
+	dw_mipi_dsi_dphy_timing_config(dsi);
+	dw_mipi_dsi_dphy_interface_config(dsi);
+
+	dw_mipi_dsi_clear_err(dsi);
+
+	mipi_dphy_init(dsi);
+	mipi_dphy_power_on(dsi);
+	dsi_write(dsi, DSI_PWR_UP, POWERUP);
+}
+
 static int dw_mipi_dsi_connector_prepare(struct rockchip_connector *conn,
                     struct display_state *state)
 {
@@ -1320,12 +1361,50 @@
     return 0;
 }

+static void dw_mipi_dsi_connector_fixup_dts(struct rockchip_connector *conn,
+                        struct display_state *state)
+{
+	struct panel_state *panel_state = &state->panel_state;
+	struct rockchip_panel *panel = panel_state->panel;
+	ofnode *rockchip_panel_node = &panel->dev->node;
+	const struct device_node *conn_node;
+	char conn_endpoint_path[128], panel_endpoint_path[128];
+	const char *conn_path, *panel_path;
+	void *blob = (void*)gd->fdt_blob;
+	ofnode node;
+	u32 phandle = 0;
+	int offset;
+
+	conn_node = ofnode_to_np(conn->dev->node);
+	conn_path = conn_node->full_name;
+	strcpy(conn_endpoint_path, conn_path);
+	strcat(conn_endpoint_path, "/ports/port@1/endpoint");
+
+	panel_path = rockchip_panel_node->np->full_name;
+	strcpy(panel_endpoint_path, panel_path);
+	strcat(panel_endpoint_path, "/ports/port@0/endpoint");
+	phandle = fdt_get_phandle(blob, fdt_path_offset(blob, panel_endpoint_path));
+	do_fixup_by_path_u32(blob, (const char*)conn_endpoint_path,
+	                "remote-endpoint", phandle, 1);
+
+	ofnode_for_each_subnode(node, conn->dev->node) {
+	    if (strstr(node.np->name, "panel") &&
+	        ofnode_valid(node) && ofnode_is_available(node) &&
+	        strcmp(node.np->full_name, panel_path)) {
+	        offset = fdt_path_offset(blob, node.np->full_name);
+	        fdt_setprop_string(blob, offset, "status", "disa");
+	    }
```

```
+   }
+}
+
 static const struct rockchip_connector_funcs dw_mipi_dsi_connector_funcs = {
     .init = dw_mipi_dsi_connector_init,
     .prepare = dw_mipi_dsi_connector_prepare,
     .unprepare = dw_mipi_dsi_connector_unprepare,
     .enable = dw_mipi_dsi_connector_enable,
     .disable = dw_mipi_dsi_connector_disable,
+   .pre_prepare = dw_mipi_dsi_connector_pre_prepare,
+   .fixup_dts = dw_mipi_dsi_connector_fixup_dts,
 };

 static int dw_mipi_dsi_probe(struct udevice *dev)
diff --git a/drivers/video/drm/dw_mipi_dsi2.c b/drivers/video/drm/dw_mipi_dsi2.c
index 4230e7c..ad4dd16 100644
--- a/drivers/video/drm/dw_mipi_dsi2.c
+++ b/drivers/video/drm/dw_mipi_dsi2.c
@@ -535,6 +535,8 @@

     if (dsi2->dsc_enable)
         grf_field_write(dsi2, IPI_FORMAT, IPI_FORMAT_DSC);
+   else
+       grf_field_write(dsi2, IPI_FORMAT, IPI_FORMAT_RGB);
 }

 static void dw_mipi_dsi2_ipi_set(struct dw_mipi_dsi2 *dsi2)
@@ -836,6 +838,9 @@
     if (!(dsi2->mode_flags & MIPI_DSI_MODE_VIDEO)) {
         conn_state->output_flags |= ROCKCHIP_OUTPUT_MIPI_DS_MODE;
         conn_state->hold_mode = dsi2->disable_hold_mode ? false : true;
+   } else {
+       conn_state->output_flags &= ~ROCKCHIP_OUTPUT_MIPI_DS_MODE;
+       conn_state->hold_mode = false;
     }

     if (dsi2->lanes > 4) {
@@ -895,6 +900,8 @@
         cstate->dsc_sink_cap.target_bits_per_pixel_x16 = 8 << 4;
         cstate->dsc_sink_cap.native_420 = 0;
         memcpy(&cstate->pps, dsi2->pps, sizeof(struct drm_dsc_picture_parameter_set));
+   } else {
+       cstate->dsc_enable = 0;
     }

     return 0;
@@ -1158,6 +1165,31 @@
         dsi2->slave->mipi_pixel_rate = dsi2->mipi_pixel_rate;
 }

+static void dw_mipi_dsi2_connector_pre_prepare(struct rockchip_connector *conn,
+                          struct display_state *state)
+{
```

```
+	struct dw_mipi_dsi2 *dsi2 = dev_get_priv(conn->dev);
+	struct connector_state *conn_state = &state->conn_state;
+	unsigned long lane_rate = 1000000000UL;
+
+	memcpy(&dsi2->mode, &conn_state->mode, sizeof(struct drm_display_mode));
+	if (dsi2->slave)
+	    memcpy(&dsi2->slave->mode, &dsi2->mode,
+	           sizeof(struct drm_display_mode));
+
+	if (dsi2->dcphy.phy)
+	    dw_mipi_dsi2_set_hs_clk(dsi2, lane_rate);
+
+	if (dsi2->slave && dsi2->slave->dcphy.phy)
+	    dw_mipi_dsi2_set_hs_clk(dsi2->slave, lane_rate);
+
+	printf("final DSI-Link bandwidth: %u %s x %d\n",
+	       dsi2->lane_hs_rate, dsi2->c_option ? "Ksps" : "Kbps",
+	       dsi2->slave ? dsi2->lanes * 2 : dsi2->lanes);
+
+	dw_mipi_dsi2_pre_enable(dsi2);
+}
+
 static int dw_mipi_dsi2_connector_prepare(struct rockchip_connector *conn,
                       struct display_state *state)
 {
@@ -1254,6 +1286,42 @@
 	return MODE_OK;
 }

+static void dw_mipi_dsi2_connector_fixup_dts(struct rockchip_connector *conn,
+                         struct display_state *state)
+{
+	struct panel_state *panel_state = &state->panel_state;
+	struct rockchip_panel *panel = panel_state->panel;
+	ofnode *rockchip_panel_node = &panel->dev->node;
+	const struct device_node *conn_node;
+	char conn_endpoint_path[128], panel_endpoint_path[128];
+	const char *conn_path, *panel_path;
+	void *blob = (void*)gd->fdt_blob;
+	ofnode node;
+	u32 phandle = 0;
+	int offset;
+
+	conn_node = ofnode_to_np(conn->dev->node);
+	conn_path = conn_node->full_name;
+	strcpy(conn_endpoint_path, conn_path);
+	strcat(conn_endpoint_path, "/ports/port@1/endpoint");
+
+	panel_path = rockchip_panel_node->np->full_name;
+	strcpy(panel_endpoint_path, panel_path);
+	strcat(panel_endpoint_path, "/ports/port@0/endpoint");
+	phandle = fdt_get_phandle(blob, fdt_path_offset(blob, panel_endpoint_path));
+	do_fixup_by_path_u32(blob, (const char*)conn_endpoint_path,
```

```
+                "remote-endpoint", phandle, 1);
+
+    ofnode_for_each_subnode(node, conn->dev->node) {
+        if (strstr(node.np->name, "panel") &&
+            ofnode_valid(node) && ofnode_is_available(node) &&
+            strcmp(node.np->full_name, panel_path)) {
+            offset = fdt_path_offset(blob, node.np->full_name);
+            fdt_setprop_string(blob, offset, "status", "disa");
+        }
+    }
+}
+
 static const struct rockchip_connector_funcs dw_mipi_dsi2_connector_funcs = {
     .pre_init = dw_mipi_dsi2_connector_pre_init,
     .init = dw_mipi_dsi2_connector_init,
@@ -1262,6 +1330,8 @@
     .enable = dw_mipi_dsi2_connector_enable,
     .disable = dw_mipi_dsi2_connector_disable,
     .mode_valid = dw_mipi_dsi2_connector_mode_valid,
+    .pre_prepare = dw_mipi_dsi2_connector_pre_prepare,
+    .fixup_dts = dw_mipi_dsi2_connector_fixup_dts,
 };

static int dw_mipi_dsi2_probe(struct udevice *dev)
diff --git a/drivers/video/drm/rockchip_connector.h b/drivers/video/drm/rockchip_connector.h
index 8c2ff75..e098b7f 100644
--- a/drivers/video/drm/rockchip_connector.h
+++ b/drivers/video/drm/rockchip_connector.h
@@ -183,6 +183,12 @@
      */
     int (*get_edid)(struct rockchip_connector *connector, struct display_state *state);
     /*
+     * for reading panel id,configure mipi host and phy simply
+     * so that reading and writing command can send succeessfully.
+     */
+    void (*pre_prepare)(struct rockchip_connector *connector, struct display_state *state);
+    void (*fixup_dts)(struct rockchip_connector *connector, struct display_state *state);
+    /*
      * call before crtc enable.
      */
     int (*prepare)(struct rockchip_connector *connector, struct display_state *state);
diff --git a/drivers/video/drm/rockchip_display.c b/drivers/video/drm/rockchip_display.c
index ae99b4d..1042f41 100644
--- a/drivers/video/drm/rockchip_display.c
+++ b/drivers/video/drm/rockchip_display.c
@@ -762,6 +762,15 @@
     if (!ret && !state->force_output)
         goto deinit;

+    if (conn->panel) {
+        rockchip_panel_pre_prepare(conn->panel);
+        if (state->enabled_at_spl == false) {
+            ret = rockchip_connector_init(state);
```

```
+           if (ret)
+               goto deinit;
+       }
+   }
+
    ret = 0;
    if (state->enabled_at_spl == true) {
 #ifdef CONFIG_SPL_BUILD
@@ -1736,12 +1745,17 @@
    int ret = 0;

    *panel = NULL;
-   panel_ofnode = dev_read_subnode(dev, "panel");
-   if (ofnode_valid(panel_ofnode) && ofnode_is_available(panel_ofnode)) {
-       ret = uclass_get_device_by_ofnode(UCLASS_PANEL, panel_ofnode,
-                         &panel_dev);
-       if (!ret)
-           goto found;
+   ofnode_for_each_subnode(panel_ofnode, dev->node) {
+       if (!strstr(panel_ofnode.np->name, "panel"))
+           continue;
+       if (ofnode_valid(panel_ofnode) &&
+           ofnode_is_available(panel_ofnode)) {
+           ret = uclass_get_device_by_ofnode(UCLASS_PANEL,
+                             panel_ofnode,
+                             &panel_dev);
+           if (!ret)
+               goto found;
+       }
    }

    ep_node = rockchip_of_graph_get_remote_node(dev->node, PORT_DIR_OUT, 0);
diff --git a/drivers/video/drm/rockchip_panel.c b/drivers/video/drm/rockchip_panel.c
index 9ddb4d8..f4b91c5 100644
--- a/drivers/video/drm/rockchip_panel.c
+++ b/drivers/video/drm/rockchip_panel.c
@@ -15,16 +15,20 @@
 #include <spi.h>
 #include <asm/gpio.h>
 #include <dm/device.h>
+#include <dm/device-internal.h>
 #include <dm/read.h>
 #include <dm/uclass.h>
 #include <dm/uclass-id.h>
 #include <linux/media-bus-format.h>
 #include <power/regulator.h>
+#include <dm/of_access.h>
+#include <fdt_support.h>

 #include "rockchip_display.h"
 #include "rockchip_crtc.h"
 #include "rockchip_connector.h"
 #include "rockchip_panel.h"
```

```
+#include "rockchip_phy.h"

 struct rockchip_cmd_header {
    u8 data_type;
@@ -42,6 +46,11 @@
    int cmd_cnt;
 };

+struct dsi_panel_id {
+   u8 *buf;
+   int len;
+};
+
 struct rockchip_panel_plat {
    bool power_invert;
    u32 bus_format;
@@ -58,6 +67,10 @@

    struct rockchip_panel_cmds *on_cmds;
    struct rockchip_panel_cmds *off_cmds;
+   struct rockchip_panel_cmds *read_id_cmds;
+
+   unsigned int id_reg;
+   struct dsi_panel_id *id;
 };

 struct rockchip_panel_priv {
@@ -411,7 +424,205 @@
    priv->enabled = false;
 }

+static int panel_simple_dsi_match_id(struct rockchip_panel *panel)
+{
+   struct rockchip_panel_plat *plat = dev_get_platdata(panel->dev);
+   struct rockchip_panel_priv *priv = dev_get_priv(panel->dev);
+   struct mipi_dsi_device *dsi = dev_get_parent_platdata(panel->dev);
+   int cmd_len = plat->id->len;
+   int read_count = 1, match_count;
+   int ret = -EINVAL;
+   u8 *panel_id;
+
+   panel_id = malloc(sizeof(char) * cmd_len);
+
+   if (priv->power_supply)
+       regulator_set_enable(priv->power_supply, !plat->power_invert);
+
+   if (dm_gpio_is_valid(&priv->enable_gpio))
+       dm_gpio_set_value(&priv->enable_gpio, 1);
+
+   if (plat->delay.prepare)
+       mdelay(plat->delay.prepare);
+
+   if (dm_gpio_is_valid(&priv->reset_gpio))
```

```
+       dm_gpio_set_value(&priv->reset_gpio, 1);
+
+   if (plat->delay.reset)
+       mdelay(plat->delay.reset);
+
+   if (dm_gpio_is_valid(&priv->reset_gpio))
+       dm_gpio_set_value(&priv->reset_gpio, 0);
+
+   if (plat->delay.init)
+       mdelay(plat->delay.init);
+
+   if (plat->read_id_cmds) {
+       rockchip_panel_send_dsi_cmds(dsi, plat->read_id_cmds);
+       free(plat->read_id_cmds);
+       plat->read_id_cmds = NULL;
+   }
+
+   if (cmd_len > 1)
+       mipi_dsi_set_maximum_return_packet_size(dsi, cmd_len);
+
+   for (int i = 0; i < read_count; i++) {
+
+       printf("panel id, reg: 0x%02x, val: ", plat->id_reg);
+       for (int j = 0; j < cmd_len; j++)
+           printf("%02x", plat->id->buf[j]);
+       printf("\n");
+       ret = mipi_dsi_dcs_read(dsi, plat->id_reg, panel_id, cmd_len);
+       if (ret > 0) {
+           printf("read panel id, reg: 0x%02x, val: ", plat->id_reg);
+           match_count = 0;
+           for (int j = 0; j < cmd_len; j++) {
+               printf("%02x", panel_id[j]);
+               if (panel_id[j] == plat->id->buf[j])
+                   match_count++;
+           }
+           printf("\n");
+           if (match_count == cmd_len) {
+               ret = 0;
+               break;
+           }
+           ret = -EINVAL;
+       } else {
+           printf("read panel timeout or no response\n");
+           ret = -ETIMEDOUT;
+       }
+   }
+
+   if (dm_gpio_is_valid(&priv->reset_gpio))
+       dm_gpio_set_value(&priv->reset_gpio, 1);
+
+   if (dm_gpio_is_valid(&priv->enable_gpio))
+       dm_gpio_set_value(&priv->enable_gpio, 0);
+
```

```
+	if (priv->power_supply)
+		regulator_set_enable(priv->power_supply, plat->power_invert);
+
+	if (plat->delay.unprepare)
+		mdelay(plat->delay.unprepare);
+
+	return ret;
+}
+
+static int panel_simple_dsi_match(struct display_state *state, int panel_num)
+{
+	struct connector_state *conn_state = &state->conn_state;
+	struct panel_state *panel_state = &state->panel_state;
+	struct rockchip_connector *conn = conn_state->connector;
+	const struct rockchip_connector_funcs *conn_funcs = conn->funcs;
+	struct rockchip_panel *panel;
+	ofnode panel_ofnode;
+	struct udevice *panel_dev;
+	int ret;
+
+	if (conn_funcs->pre_init) {
+		ret = conn_funcs->pre_init(conn, state);
+		if (ret) {
+			printf("failed to pre_init connector\n");
+			goto deinit;
+		}
+	}
+
+	if (conn_funcs->init) {
+		ret = conn_funcs->init(conn, state);
+		if (ret) {
+			printf("failed to init connector\n");
+			goto deinit;
+		}
+	}
+
+	if (conn->phy)
+		rockchip_phy_init(conn->phy);
+
+	if (conn_funcs->pre_prepare)
+		conn_funcs->pre_prepare(conn, state);
+
+	ofnode_for_each_subnode(panel_ofnode, conn->dev->node) {
+		if (!strstr(panel_ofnode.np->name, "panel"))
+			continue;
+		if (ofnode_valid(panel_ofnode) &&
+			ofnode_is_available(panel_ofnode)) {
+			ret = uclass_get_device_by_ofnode(UCLASS_PANEL,
+							  panel_ofnode,
+							  &panel_dev);
+			if (ret) {
+				printf("this sub node is not a panel node\n");
+				continue;
```

```
+           }
+       } else {
+           continue;
+       }
+
+       panel = (struct rockchip_panel *)dev_get_driver_data(panel_dev);
+       if (!panel) {
+           printf("can not found panel\n");
+           ret = -ENODEV;
+           break;
+       }
+
+       panel_state->panel = panel;
+       conn->panel = panel;
+       panel->state = state;
+
+       printf("try to match %s\n", panel_ofnode.np->full_name);
+       ret = panel_simple_dsi_match_id(panel);
+       if (!ret)
+           break;
+
+       if (panel_num > 1) {
+           device_remove(panel_dev, DM_REMOVE_NORMAL);
+           panel_num--;
+       }
+   }
+
+   if (conn_funcs->disable)
+       conn_funcs->disable(conn, state);
+
+   if (conn_funcs->unprepare)
+       conn_funcs->unprepare(conn, state);
+
+   if (conn_funcs->fixup_dts)
+       conn_funcs->fixup_dts(conn, state);
+
+deinit:
+   if (conn_funcs->deinit)
+       conn_funcs->deinit(conn, state);
+
+   return ret;
+}
+
+static void panel_simple_pre_prepare(struct rockchip_panel *panel)
+{
+   struct display_state *state = panel->state;
+   struct connector_state *conn_state = &state->conn_state;
+   struct rockchip_connector *conn = conn_state->connector;
+   char *conn_name = (char *)conn->dev->node.np->full_name;
+   ofnode panel_ofnode;
+   int panel_num = 0;
+
+   conn_state->bus_format = panel->bus_format;
```

```
+
+   ofnode_for_each_subnode(panel_ofnode, conn->dev->node) {
+       if (strstr(panel_ofnode.np->name, "panel") &&
+           ofnode_valid(panel_ofnode) &&
+           ofnode_is_available(panel_ofnode))
+           panel_num++;
+   }
+
+   if (panel_num > 1 && strstr(conn_name, "dsi")) {
+       if (panel_simple_dsi_match(state, panel_num))
+           printf("fail to match valid panel\n");
+       else
+           printf("success to match valid panel\n");
+   }
+}
+
 static const struct rockchip_panel_funcs rockchip_panel_funcs = {
+   .pre_prepare = panel_simple_pre_prepare,
    .prepare = panel_simple_prepare,
    .unprepare = panel_simple_unprepare,
    .enable = panel_simple_enable,
@@ -436,6 +647,17 @@

    plat->bus_format = dev_read_u32_default(dev, "bus-format",
                       MEDIA_BUS_FMT_RBG888_1X24);
+   plat->id_reg =  dev_read_u32_default(dev, "id-reg", 0);
+   data = dev_read_prop(dev, "id", &len);
+
+   if (data) {
+       plat->id = malloc(sizeof(*plat->id));
+       if (!plat->id)
+           return -ENOMEM;
+       plat->id->buf = malloc(sizeof(char) * len);
+       memcpy(plat->id->buf, data, len);
+       plat->id->len = len;
+   }
    plat->bpc = dev_read_u32_default(dev, "bpc", 8);

    data = dev_read_prop(dev, "panel-init-sequence", &len);
@@ -451,6 +673,19 @@
        }
    }

+   data = dev_read_prop(dev, "panel-read-id-sequence", &len);
+   if (data) {
+       plat->read_id_cmds = calloc(1, sizeof(*plat->read_id_cmds));
+       if (!plat->read_id_cmds)
+           return -ENOMEM;
+
+       ret = rockchip_panel_parse_cmds(data, len, plat->read_id_cmds);
+       if (ret) {
+           printf("failed to parse panel read id sequence\n");
+           goto free_read_id_cmds;
```

```
+       }
+   }
+
    data = dev_read_prop(dev, "panel-exit-sequence", &len);
    if (data) {
        plat->off_cmds = calloc(1, sizeof(*plat->off_cmds));
@@ -470,6 +705,8 @@

 free_cmds:
    free(plat->off_cmds);
+free_read_id_cmds:
+   free(plat->read_id_cmds);
 free_on_cmds:
    free(plat->on_cmds);
    return ret;
@@ -574,6 +811,19 @@
    return 0;
 }

+static int panel_simple_remove(struct udevice *dev) {
+   struct rockchip_panel_priv *priv = dev_get_priv(dev);
+
+   printf("remove %s\n", dev->node.np->full_name);
+
+   if (dm_gpio_is_valid(&priv->enable_gpio))
+       dm_gpio_free(dev, &priv->enable_gpio);
+   if (dm_gpio_is_valid(&priv->reset_gpio))
+       dm_gpio_free(dev, &priv->reset_gpio);
+
+   return 0;
+}
+
 static const struct udevice_id rockchip_panel_ids[] = {
    { .compatible = "simple-panel", },
    { .compatible = "simple-panel-dsi", },
@@ -587,6 +837,7 @@
    .of_match = rockchip_panel_ids,
    .ofdata_to_platdata = rockchip_panel_ofdata_to_platdata,
    .probe = rockchip_panel_probe,
+   .remove = panel_simple_remove,
    .priv_auto_alloc_size = sizeof(struct rockchip_panel_priv),
    .platdata_auto_alloc_size = sizeof(struct rockchip_panel_plat),
 };
diff --git a/drivers/video/drm/rockchip_panel.h b/drivers/video/drm/rockchip_panel.h
index 5e49fe3..b503e84 100644
--- a/drivers/video/drm/rockchip_panel.h
+++ b/drivers/video/drm/rockchip_panel.h
@@ -12,10 +12,12 @@
 struct rockchip_conn;

 struct rockchip_panel_funcs {
+   void (*pre_prepare)(struct rockchip_panel *panel);
    void (*prepare)(struct rockchip_panel *panel);
```

```
    void (*unprepare)(struct rockchip_panel *panel);
    void (*enable)(struct rockchip_panel *panel);
    void (*disable)(struct rockchip_panel *panel);
+   void (*init)(struct rockchip_panel *panel);
    int (*get_mode)(struct rockchip_panel *panel,
            struct drm_display_mode *mode);
 };
@@ -48,6 +50,15 @@
        state->conn_state.bpc = panel->bpc;
 }

+static inline void rockchip_panel_pre_prepare(struct rockchip_panel *panel)
+{
+   if (!panel)
+       return;
+
+   if (panel->funcs && panel->funcs->pre_prepare)
+       panel->funcs->pre_prepare(panel);
+}
+
 static inline void rockchip_panel_prepare(struct rockchip_panel *panel)
 {
    if (!panel)
```

3. 调试

3.1 关键日志分析

根据如下去排查屏端获取 ID 失败的原因：

```
# ID匹配成功日志
read panel id, reg: 0xDA, val: 09       # 读取到的ID值
success to match valid panel           # 驱动确认匹配

# 异常场景
fail to read panel id                   # 通信失败
ID mismatch: expect 08 got 09          # ID不匹配
```

3.2 FDT调试

```
adb pull /sys/firmware/fdt ./fdt

在 kernel 根目录下执行 DTB 反编译
./scripts/dtc/dtc  -I dtb -O dts -o fdt.dts ./fdt
```

查看 fdt.dts 验证节点状态是否符合预期，比如：

```
panel0@0 { status = "okay"; ... };
panel1@0 { status = "disabled"; ... };
```